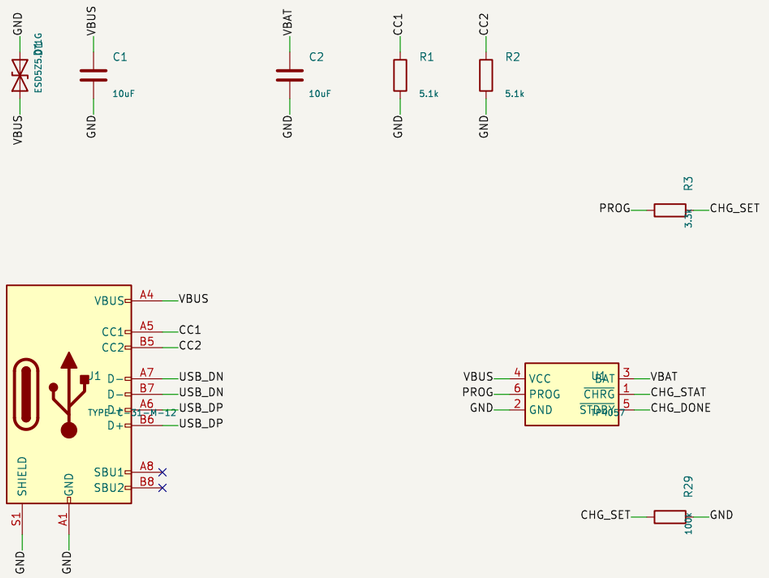
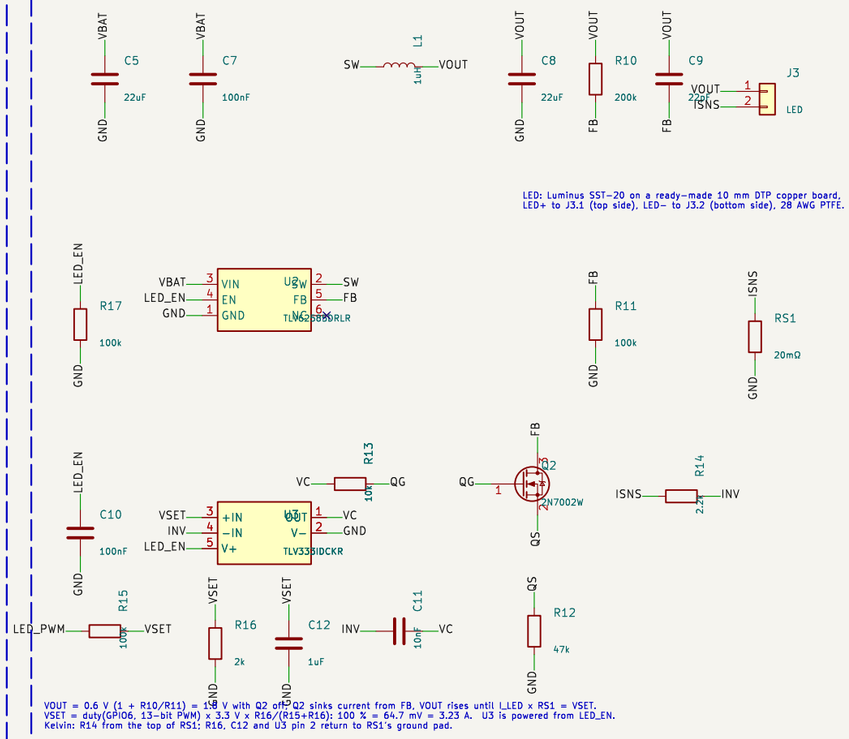
Blocks 1–2: USB-C (power + data), TP4057 charger with charge pause



USB-C: Rd 5.1 k on CC1 and CC2 (5 V sink). D+/D− go to the ESP32-C3 native USB (flashing, console).
TP4057: R3 3.3 k returns PROG to CHG_SET (GPIO20): low = approx. 280 mA, floating = 10 mA trickle via R29, high = charger off (only while USB is present). CHRG/STDBY are read by GPIO4/GPIO5 (internal pull-ups).

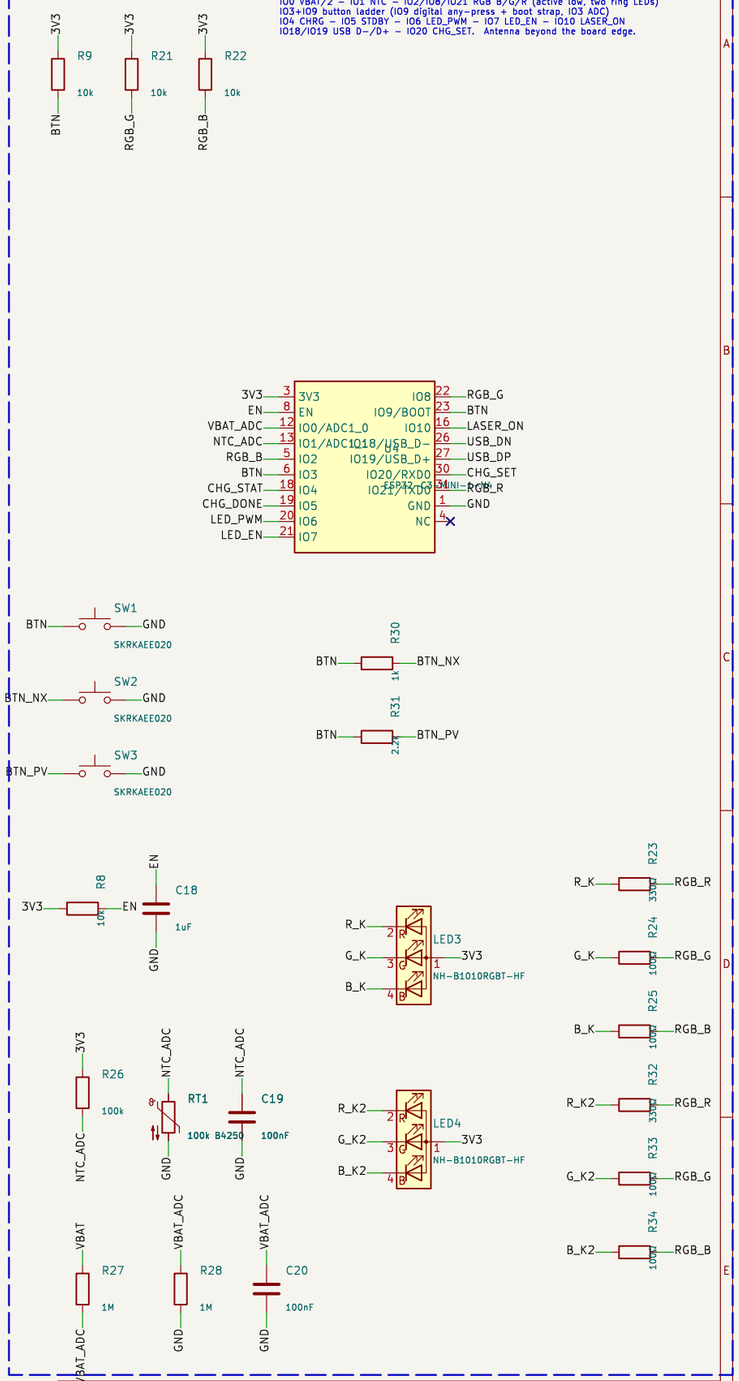
Block 4: synchronous buck LED driver with cascaded current loop



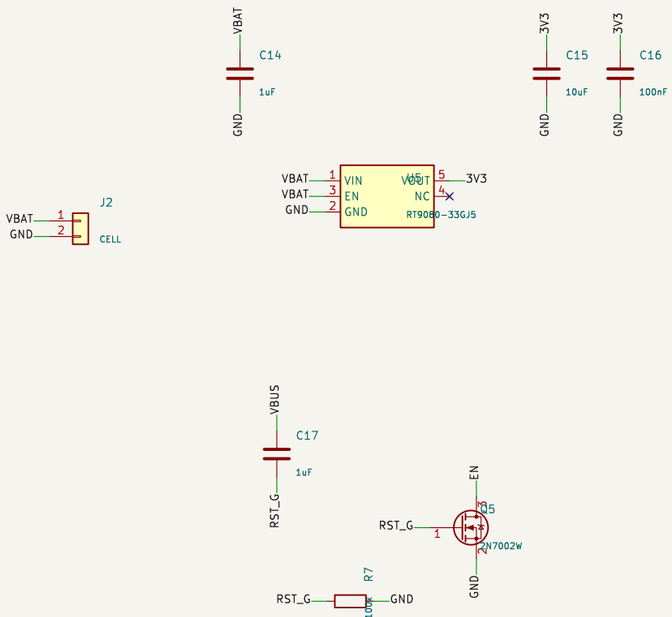
VOUT = 0.6 V (1 + R10/R11) = 1.8 V with Q2 off. Q2 sinks current from FB, VOUT rises until I_LED x RS1 = VSET.
VSET = duty(GPIO6, 13-bit PWM) x 3.3 V x R16/(R15+R16): 100 % = 64.7 mV = 3.23 A. U3 is powered from LED_EN.
Kelvin: R14 from the top of RS1; R16, C12 and U3 pin 2 return to RS1's ground pad.

Block 6: ESP32-C3-MINI-1 (BLE, USB), 3-button ladder, RGB ring LEDs, NTC, VBAT sense

GPIO map:
IO0 VBAT/2 — IO1 NTC — IO2/IO8/IO21 RGB B/G/R (active low, two ring LEDs)
IO3+IO9 button ladder (IO9 digital any-press + boot strap, IO3 ADC)
IO4 CHRG — IO5 STDBY — IO6 LED_PWM — IO7 LED_EN — IO10 LASER_ON
IO18/IO19 USB D−/D+ — IO20 CHG_SET. Antenna beyond the board edge.

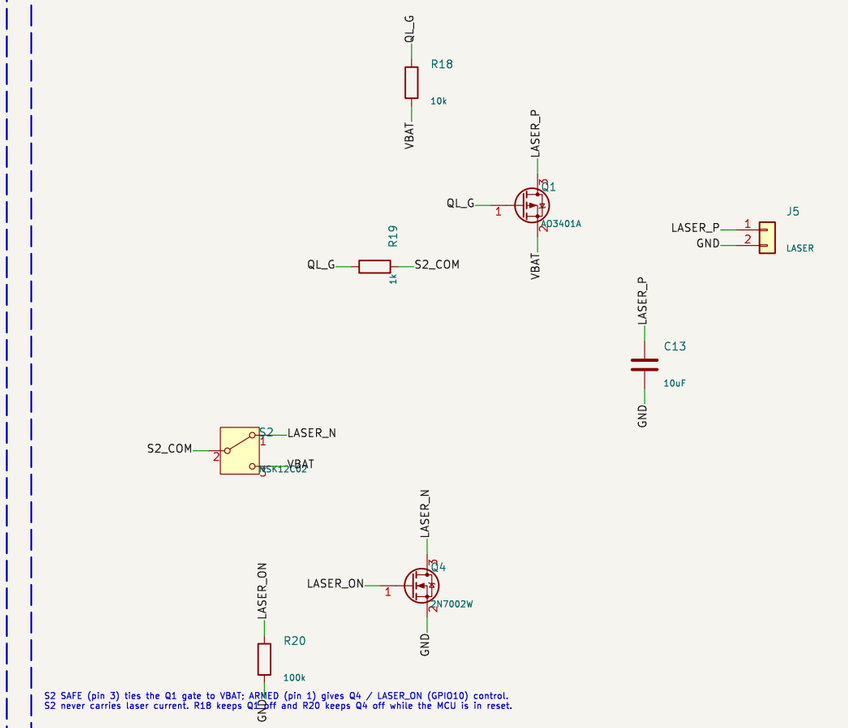


Block 3: cell pads, 3.3 V LDO, VBUS-edge reset



J2: two LP601646 cells in parallel (two wires per pad): red = BAT+ (pin 1), black = BAT− (pin 2).
U5 RT9080: 3.3 V, EN tied to VIN, 2 uA quiescent. Q5/C17/R7: plugging in USB pulls EN low for approx. 0.1 s, so the module restarts; holding the button (GPIO9 strap) while plugging in starts the ROM download mode.

Block 5: laser load switch with hardware interlock (gate path)



S2 SAFE (pin 3) ties the Q1 gate to VBAT; ARMED (pin 1) gives Q4 / LASER_ON (GPIO10) control.
S2 never carries laser current. R18 keeps Q1 off and R20 keeps Q4 off while the MCU is in reset.

KiCad 7 project; JLCPCB fabrication and assembly files in production/
Generated from tools/kicad/flashka_parts.py — nets connect by label

LuxLibre

Sheet: /
File: FlashKa_main.kicad_sch

Title: FlashKa Keychain v0.6 — main PCB

Size: A3	Date: 2026-10-16	**Rev: v0.6**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1